



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

- We used SpaceX API and BeautifulSoup to collect data, performed ETL using SQL, data visualization, including a dashboard, and geoanalysis using folium.
- We performed data wrangling by cleaning the data and feature engineering.
- We selected the best model to predict the outcome of a Falcon 9 launch.
- After careful comparison, we use a Decision Tree classifier with hyperparameter tuning, yielding a model with 89% accuracy.

Introduction

- In this project, we will develop a model to predict if the Falcon 9 stage of SpaceX will land successfully.
- By predicting the outcome of the landing, and therefore the cost of a launch, we can bid against SpaceX for a rocket launch.
- This implies exploring the data in detail to identify the main features and the relationship between them and the launch outcome.
- For the predictive modeling, we aim to compare four different models, perform grid search for hyperparameter tuning and select the best performer.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Data is retrieved via SpaceX API and web scraping with BeautifulSoup, stored as a Dataframe, cleaned and specialized to the Falcon 9 launches.
- Perform data wrangling
 - Data is cleaned, separated into numerical and categorical columns and classified according to the mission outcomes.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

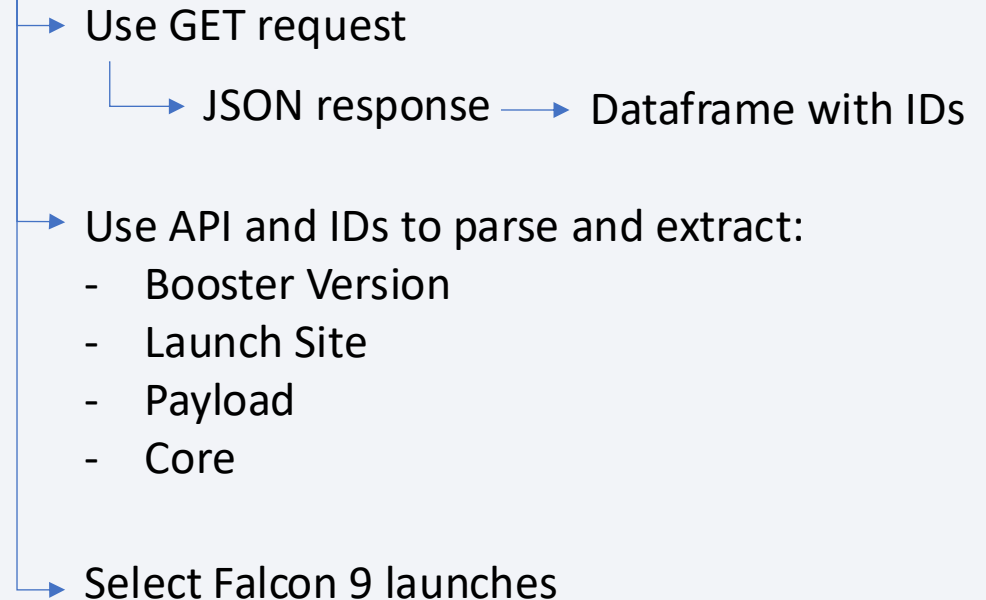
Data Collection

- Data sets collected via
 - SpaceX API: JSON response
 - Web scraping: HTML response
- Final pandas Dataframe with the parsed data

Data Collection – SpaceX API

- After launching the SpaceX API, we extract the IDs of the rocket launches.
- We use the IDs to extract the properties of each launch.
- We build a Dataframe with the data and filter the Falcon 9 launches.
- The file can be found [here](#).

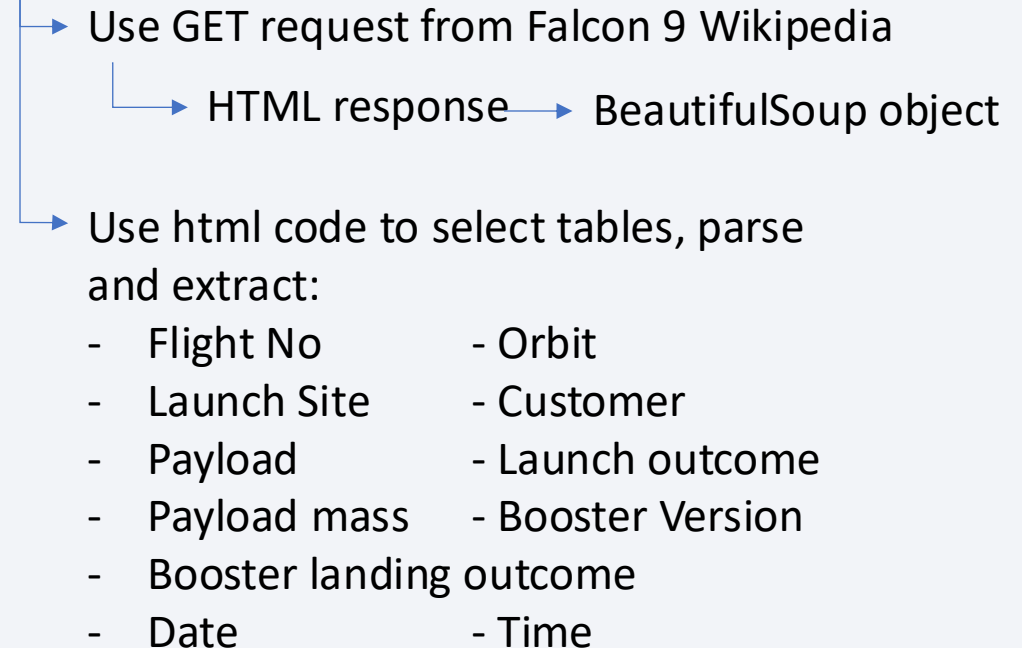
Launch SpaceX API



Data Collection - Scrapping

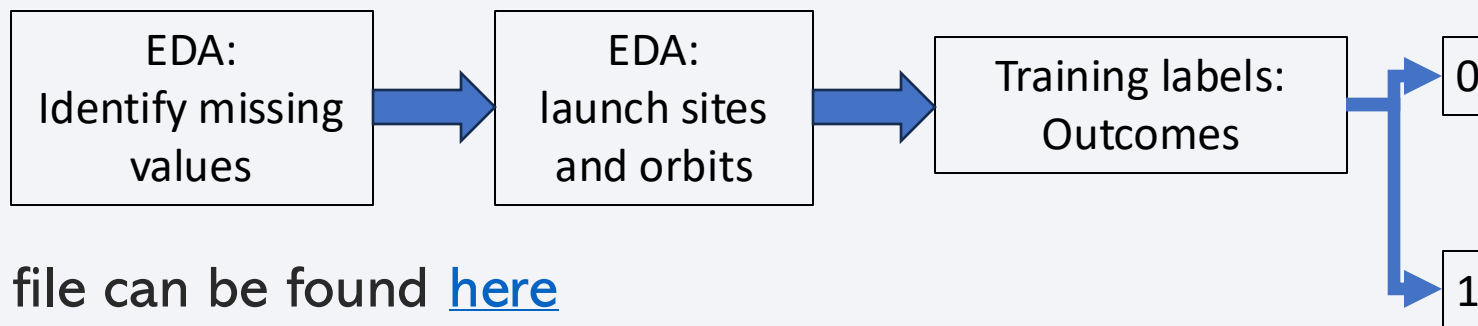
- We use the [Wikipedia page](#) for Falcon 9 launches.
- Using BeautifulSoup we parse the html and extract the relevant columns.
- The file can be found [here](#).

Web scraping

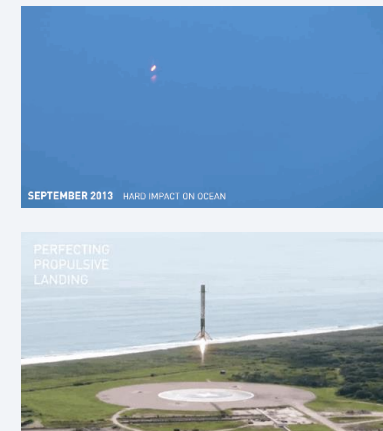


Data Wrangling

- First, we identify missing values in the landing pad column, as well as identify data types.
- Afterwards, we explore the launches sites and orbit types of each launch, and find that most of the rockets are launched at CCAFS SLC 40, and to a GTO orbit.
- Using the outcomes, we identify the successful and unsuccessful landings, and transform their values into numerical Booleans (0/1) for feature engineering.



- The file can be found [here](#)



EDA with Data Visualization

- We plotted the following relationships due to their possible correlation with the landing outcome:
 - Payload mass, launch site, and flight number.
 - Success rate, orbit type, payload mass, and flight number.
 - Launch success yearly trend.
- The file can be found [here](#)

EDA with SQL

- The SQL queries we performed were to obtain:
 - The names of the unique launch sites.
 - 5 records where launch sites begin with 'CCA'
 - The total payload mass carried by boosters launched by NASA (CRS)
 - The average payload mass carried by booster version F9 v1.1.
 - The date when the first successful landing outcome in a ground pad was archived.
 - The names of the boosters which succeeded in a drone ship and have a payload mass between 4 and 6 tones.
 - Total number of successful and failure mission outcomes
 - The names of the booster versions that carried the maximum payload mass
 - The month names, failure landing outcomes in a drone ship, booster versions and launch sites for the year 2015.
 - The ranked count of landing outcomes between the data 04/06/2010 and 20/03/2017.
- The file can be found [here](#)

Build an Interactive Map with Folium

- We created a Folium map centered at NASA Johnson Space Center (NJSC) at Houston, TX. Using the map objects circle and marker, we added a circle around NJSC and the other launch sites within the data set, and a label (marker) displaying the name of the site.
- Within each launch, we added a marker cluster object to show the outcome of the landing (success/failure), indicated by the color in the customized icons.
- Finally, we study the proximity of the launch sites to cities, railways, highways and costal lines by drawing a PolyLine and computing the distance to each item using MousePosition.
- The file can be found [here](#)

Build a Dashboard with Plotly Dash

- After selecting a launch site or all of them using a dropdown menu, we made a dashboard displaying:
 - Pie chart with total success landings corresponding to all sites
 - Pie chart with success rate specialized to a specific site.
- Additionally, we plot the successes and failures for the different booster version categories, as a function of the payload mass, which range is controlled by a slider.
- By studying these plots, we can study the optimum site launch, booster category version and payload mass in order to have a successful outcome.
- The file can be found [here](#)

Predictive Analysis (Classification)

- We identify the target variable as class, this is, the outcome of the landing.
- After standardizing the data, we split it into train and test data, with the test size being a 20% of the total.
- Then, we performed a grid search to look for the best parameters for:
 - Logistic regression
 - Support vector machine
 - Decision tree
 - K-nearest neighbors
- And evaluated their scores via:
 - Accuracy on train data
 - Accuracy on test data
 - Confusion matrix
- The file can be found [here](#)

Results

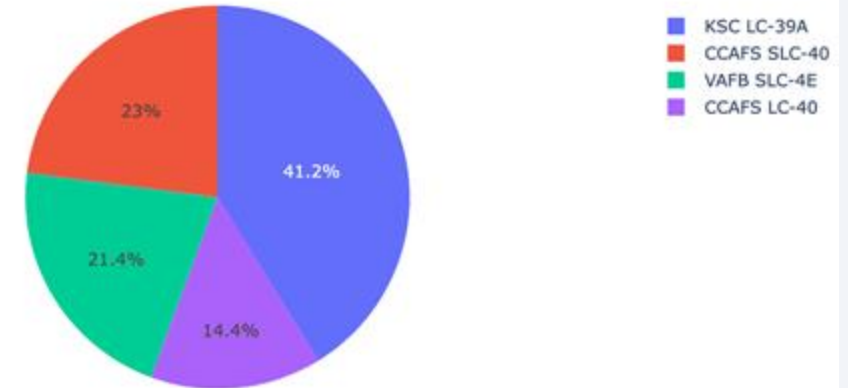
EDA Visualization

- The outcome is skewed within each launch site
- Most of the failed launches are in CCAFS SLC 40
- KSLC LC 39A has the higher success rate
- Launches with heavier payload masses are mostly successful
- The most successful orbits are ES-L1, GEO, HEO and SSO.
- The success in the LEO orbit seems to be related to the number of flights, although is not the case for the GTO orbit
- Heavy payloads are more successful in the LEO, ISS and PO orbits.
- The success rate has been increasing since 2013.

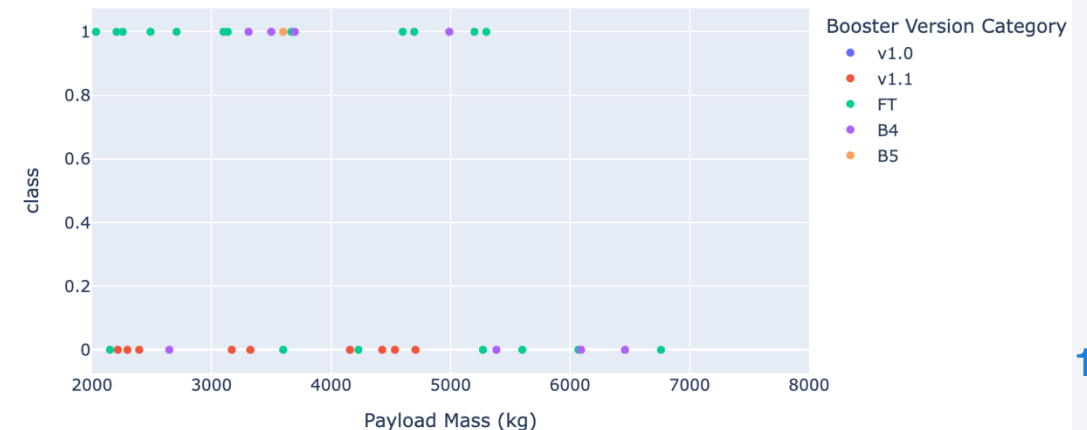
EDA SQL

- Total payload mass carried by boosters launched by NASA (CRS) is 48213 Kg
- The average payload mass carried by booster version F9 v1.1 is 2534.66 Kg
- The first successful landing outcome in ground pad was archived on the 22/12/2015.
- There is a total of 61 successful missions and 10 failed missions.
- The most successful orbits are ES-L1, GEO, HEO and SSO.
- The most landing outcomes are using a drone ship, and the least are the parachute landing.

Total success launches by site

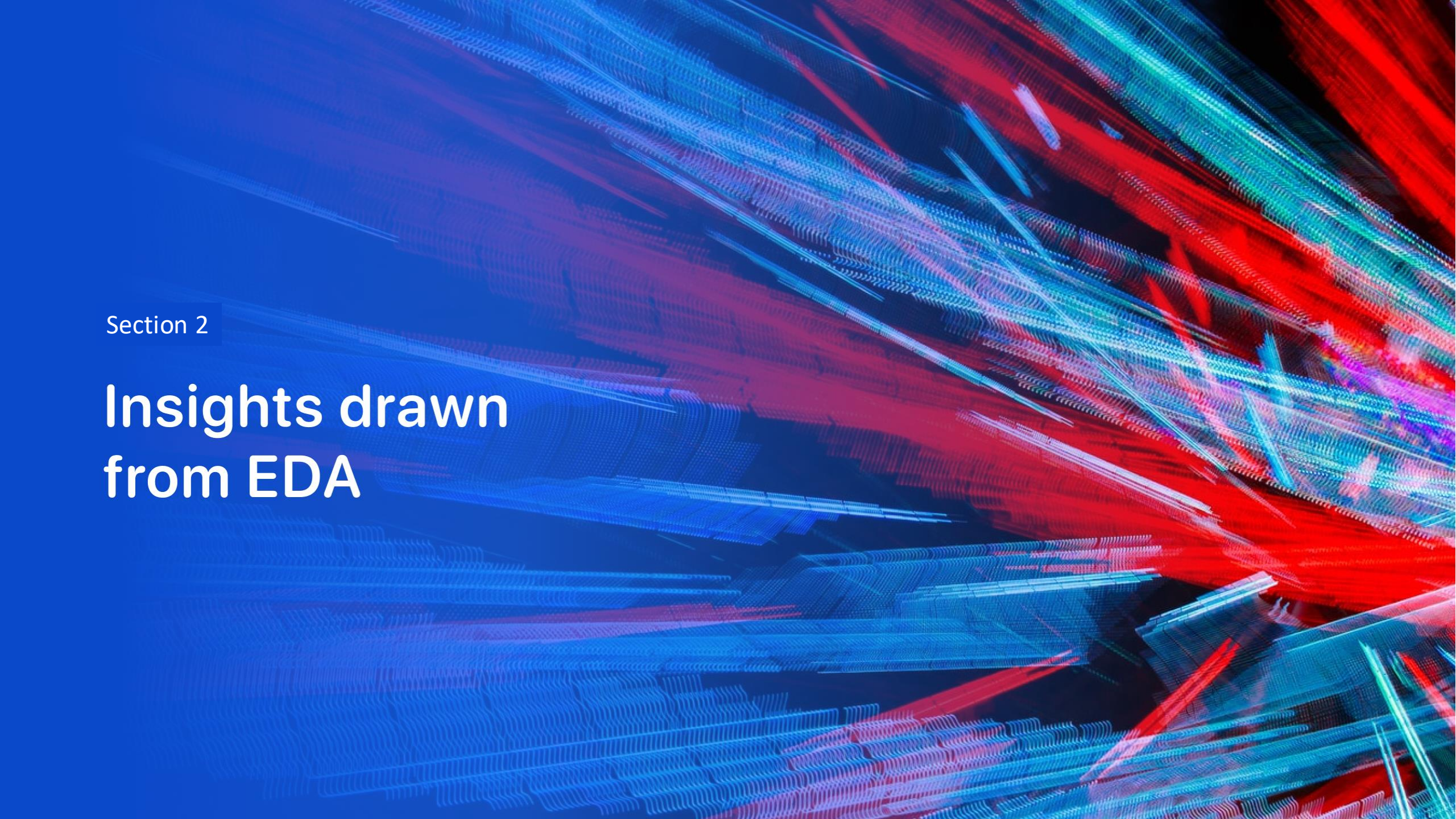


Outcome vs payload mass by booster version



Results

- The predictive analysis revealed that the best model to predict the landing outcomes is a Decision Tree with:
 - Criterion: 'gini'
 - Max_depth: 6
 - Max_features='sqrt'
 - Min_samples_split: 10
 - Splitter: random
- Overall reaching the most accuracy of all models with a value of 0.89.

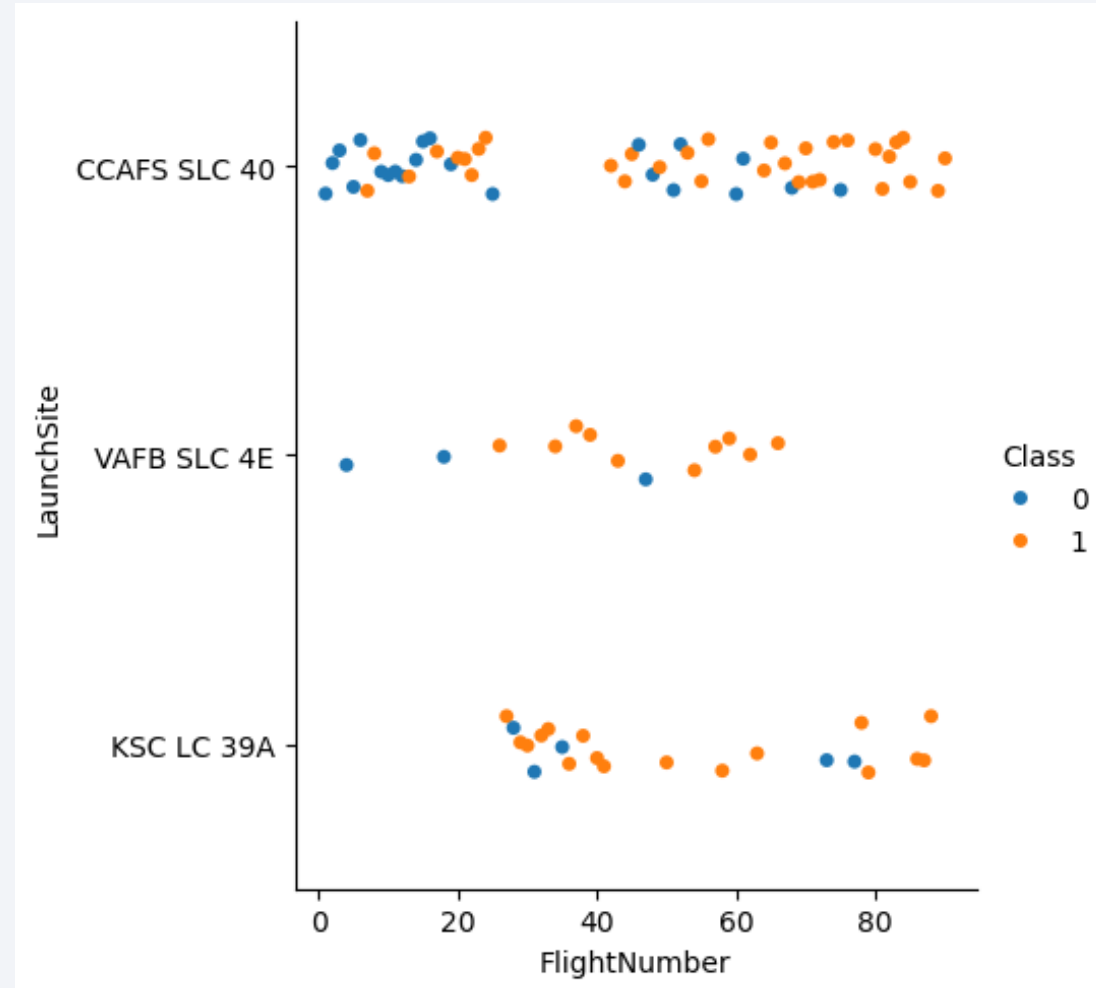
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

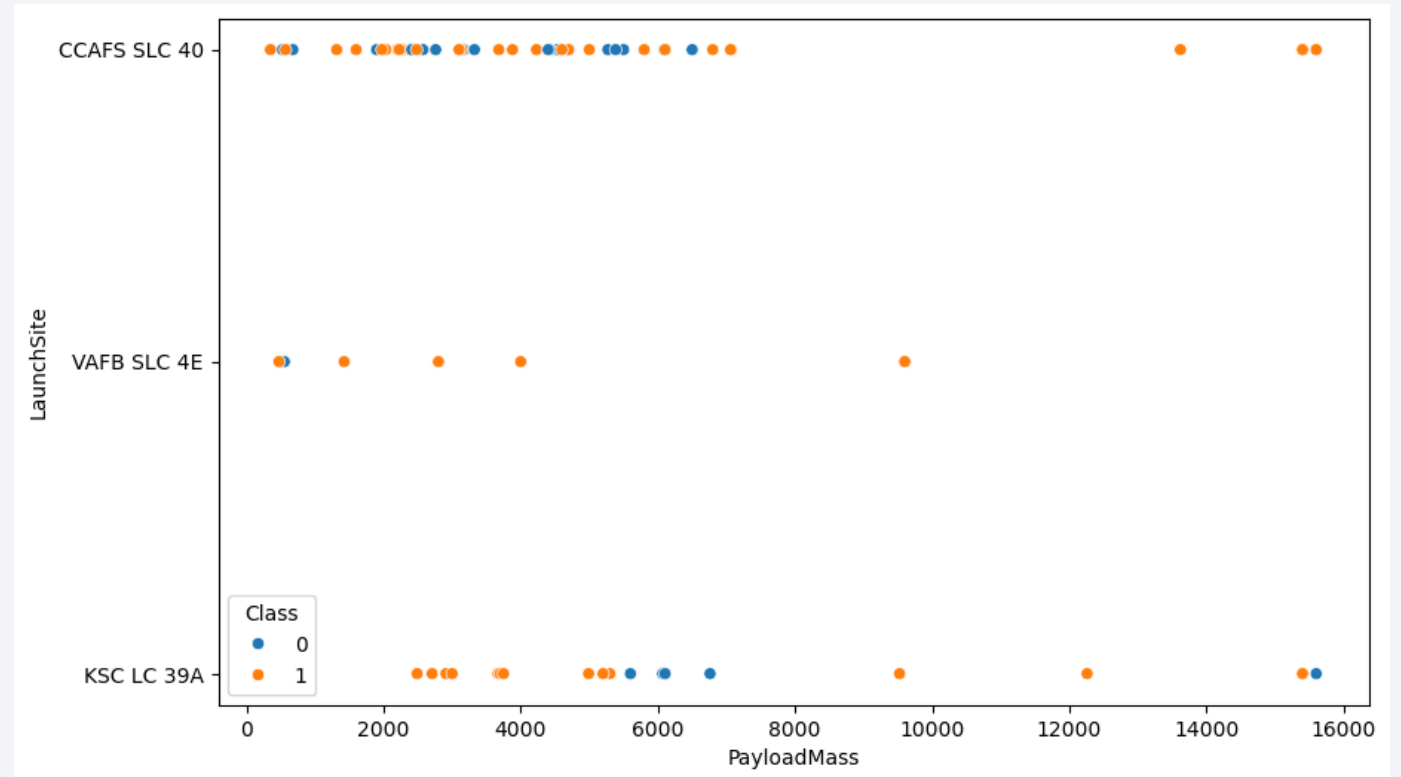
Flight Number vs. Launch Site

- CCAFS SLC 40 has the greatest number of landing outcomes, with higher success rate towards higher number of flights.
- VAFB SLC 4E has the least amount of data, with mostly successful landings.
- KSC LC 39A has mostly successful landings, and higher number of flights than VAFB SLC 4E.



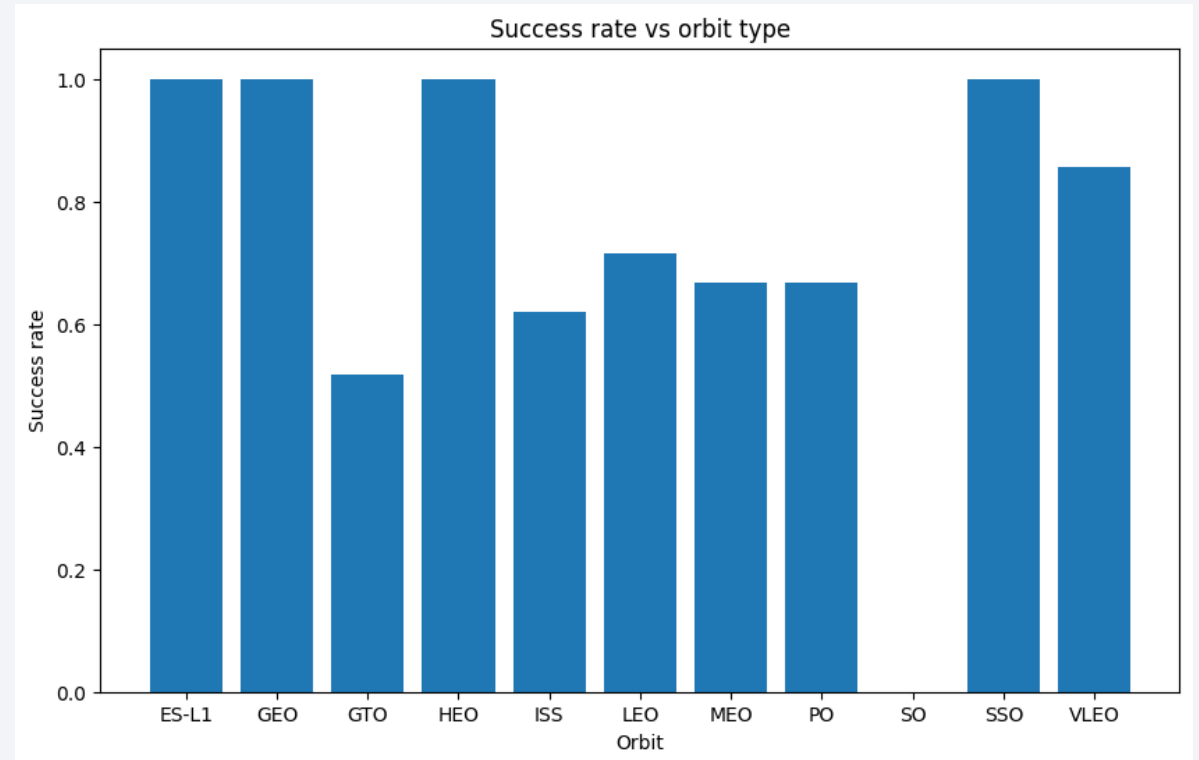
Payload vs. Launch Site

- CCAFS SLC 40 has a payload mass distribution mostly between 0 and 8k with a few successful outliers between 12k and 16k.
- VAFB SLC 4E successful landings increase with the payload mass.
- KSC LC 39A has most successful landings for a payload mass lighter than 6k Kg, but no signs of an overall apparent distribution.



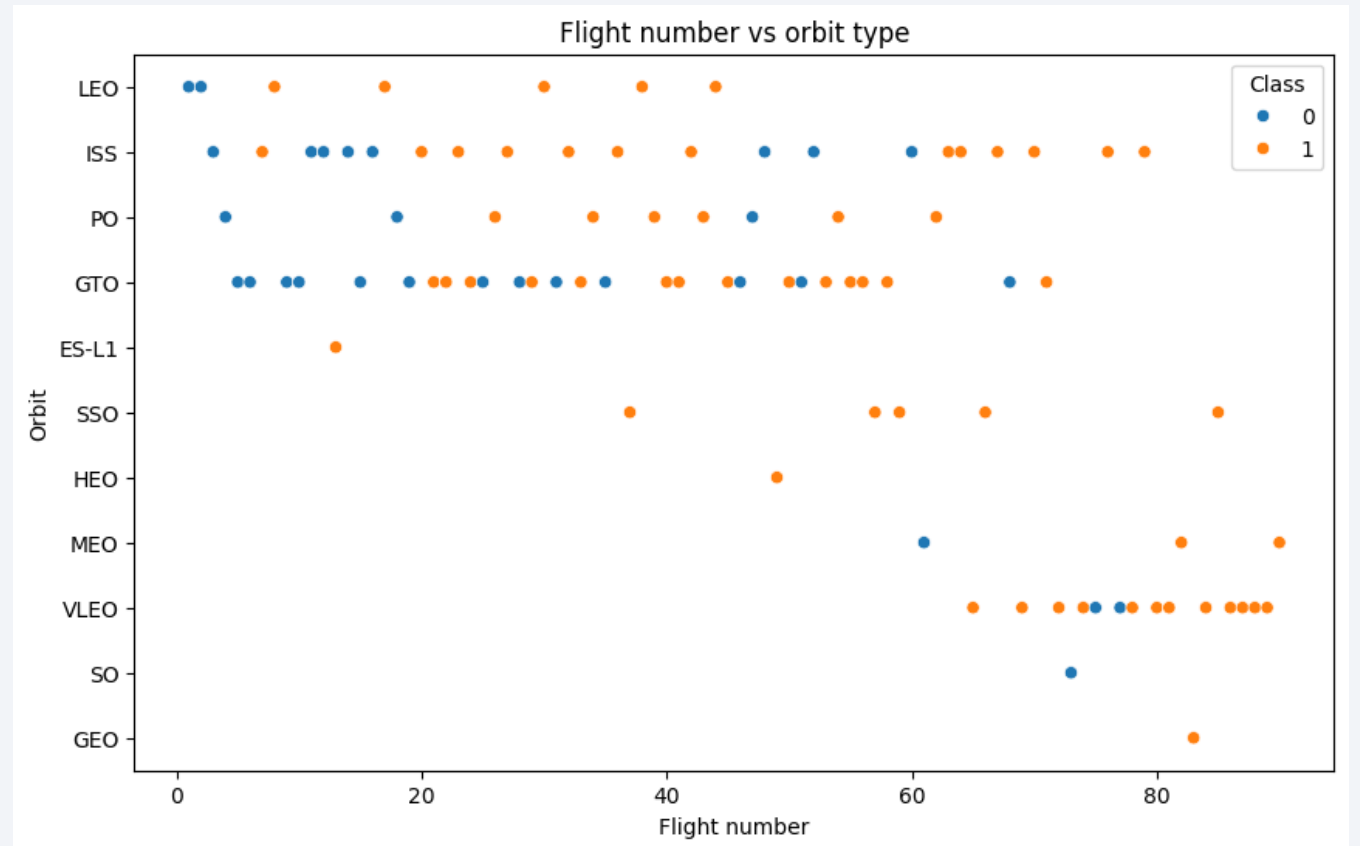
Success Rate vs. Orbit Type

- The most successful orbits are:
 - ES-L1
 - GEO
 - HEO
 - SSO
- The most unsuccessful one is SO.
- We have to keep in mind that there is only one data point for the orbits HEO, ES-L1, SO and GEO.



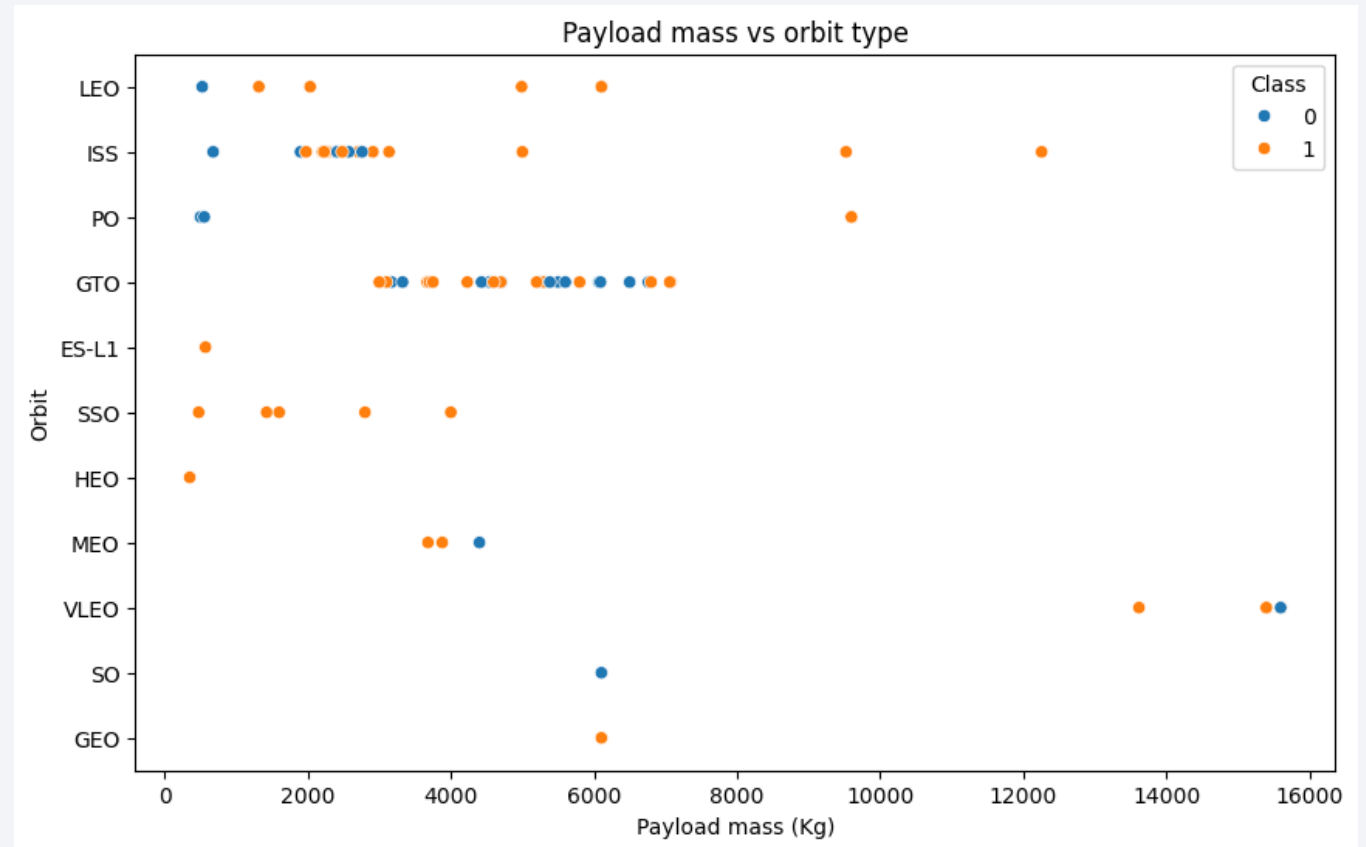
Flight Number vs. Orbit Type

- For the LEO orbit, all landings are successful after more than 10 flights.
- It seems to be a similar case for the SSO orbit.
- The GTO orbit has the most data points, and it shows no relationship with the number of flights.
- The VLEO orbit has mostly success landings as the number of flights increases, except for two flights.



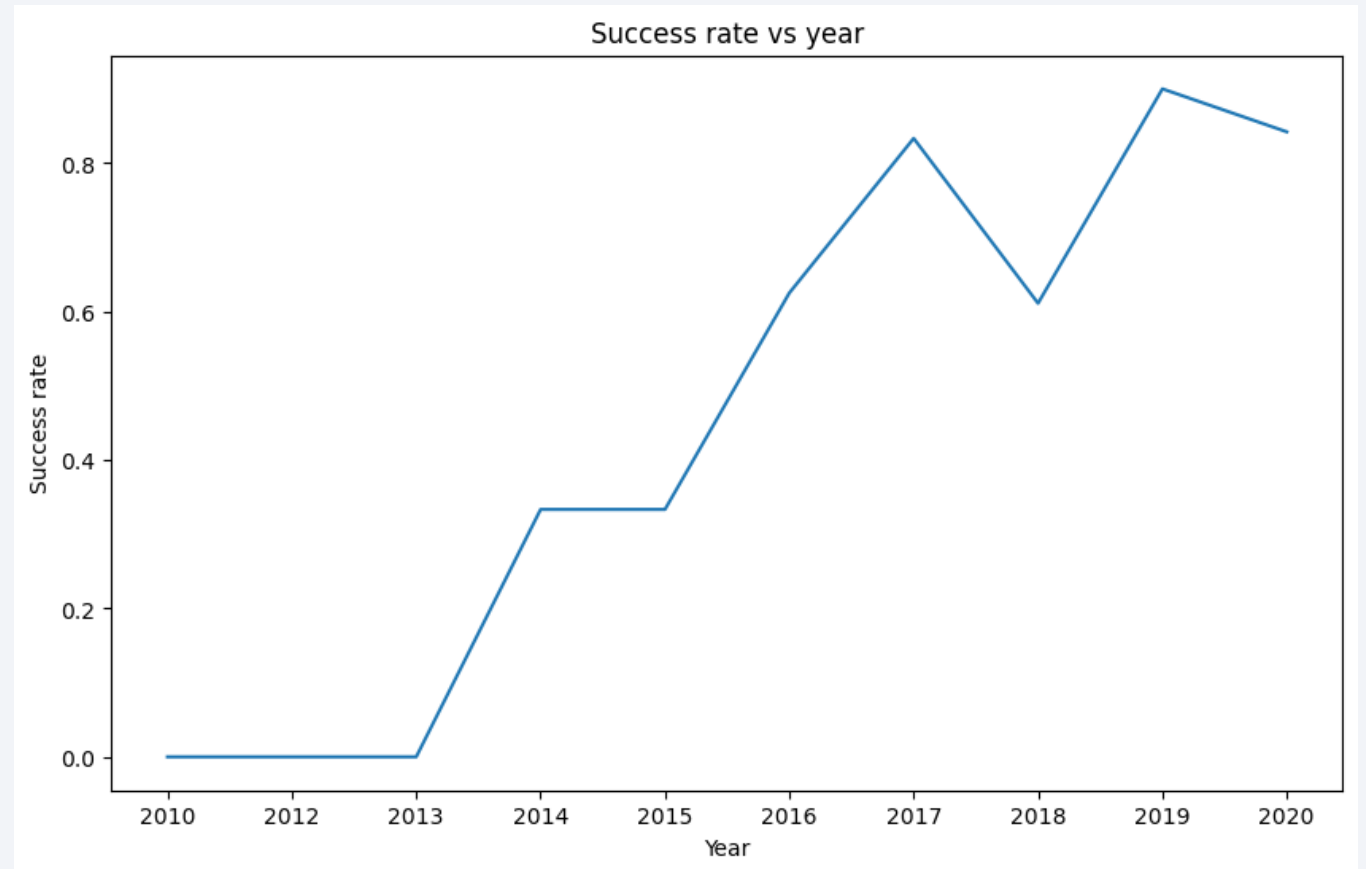
Payload vs. Orbit Type

- Heavier loads result in a successful landing for the LEO, ISS and PO orbits
- The SSO orbit has a 100% success rate for the payload masses under 5k Kg.
- The GTO orbit does not seem to follow any trends.



Launch Success Yearly Trend

- The success rate improves over time, with the exception of 2018
- A quick Google search indicates that it corresponds to problems with the hydraulic systems and the landing leg deployment.



All Launch Site Names

- SQL query to select the names of the launch sites:
 - `select distinct Launch_Site from SPACEXTABLE`
- There are four different launch sites

Launch site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- SQL query to select 5 records where launch sites begin with the string 'CCA':
 - select * from SPACEXTABLE where Launch_Site like '%CCA%' limit 5
- We find the following records:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- SQL query to calculate the total payload carried by boosters from NASA (CRS)
 - `select sum(PAYLOAD_MASS__KG_) from SPACEXTABLE where Customer like '%CRS%'`
- We find a total payload mass of 48213 Kg.

Average Payload Mass by F9 v1.1

- SQL query to calculate the average payload mass carried by booster version F9 v1.1
 - `select avg(PAYLOAD_MASS__KG_) from SPACEXTABLE where Booster_Version like '%F9 v1.1%'`
- We find an average payload mass of 2534.67 Kg.

First Successful Ground Landing Date

- SQL query to find the dates of the first successful landing outcome on ground pad
 - `select min(Date) from SPACEXTABLE where Landing_Outcome like '%Success (ground pad)%'`
- We find that the first successful landing outcome on ground pad was on 22/12/2015

Successful Drone Ship Landing with Payload between 4000 and 6000

- SQL query to list the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
 - `select Booster_Version from SPACEXTABLE where Landing_Outcome like '%Success (drone ship)%' and PAYLOAD_MASS__KG_ BETWEEN 4000 and 6000`
- We find the following boosters:

Booster version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- SQL query to calculate the total number of successful and failure mission outcomes
 - `select (select count(Landing_Outcome) from SPACEXTABLE where Landing_Outcome like '%Success%') as SuccessCount, (select count(Landing_Outcome) from SPACEXTABLE where Landing_Outcome like '%Failure%') as FailureCount`
- We find the following count:

Success	Failure
61	10

Boosters Carried Maximum Payload

- SQL query to list the names of the booster which have carried the maximum payload mass
 - `select Booster_Version from SPACEXTABLE where PAYLOAD_MASS_KG_ = (select max(PAYLOAD_MASS_KG_) from SPACEXTABLE)`
- We find the booster versions displayed in the table.

Booster version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- SQL query to list the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
 - `select substr(Date,6,2) as month, Landing_Outcome, Booster_Version, Launch_Site from SPACEXTABLE where substr(Date,0,5)='2015' and Landing_Outcome like '%Failure (drone ship)%'`
- We find that all failed landings in 2015 were launched from CCAFS LC-40 in a drone ship in January and April.

Month	Landing outcome	Booster version	Launch site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- SQL query to rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
 - `select Landing_Outcome, count(Landing_Outcome) as count from SPACEXTABLE where Date between '2010-06-04' and '2017-02-20' group by Landing_Outcome order by count desc`
- We find that the higher number of counts corresponds to a launch without attempt to land, whereas the second higher count corresponds to both successful and failed drone ship landings.
- The smaller count corresponds to failed landings with parachute, and precluded landings with drone ship.

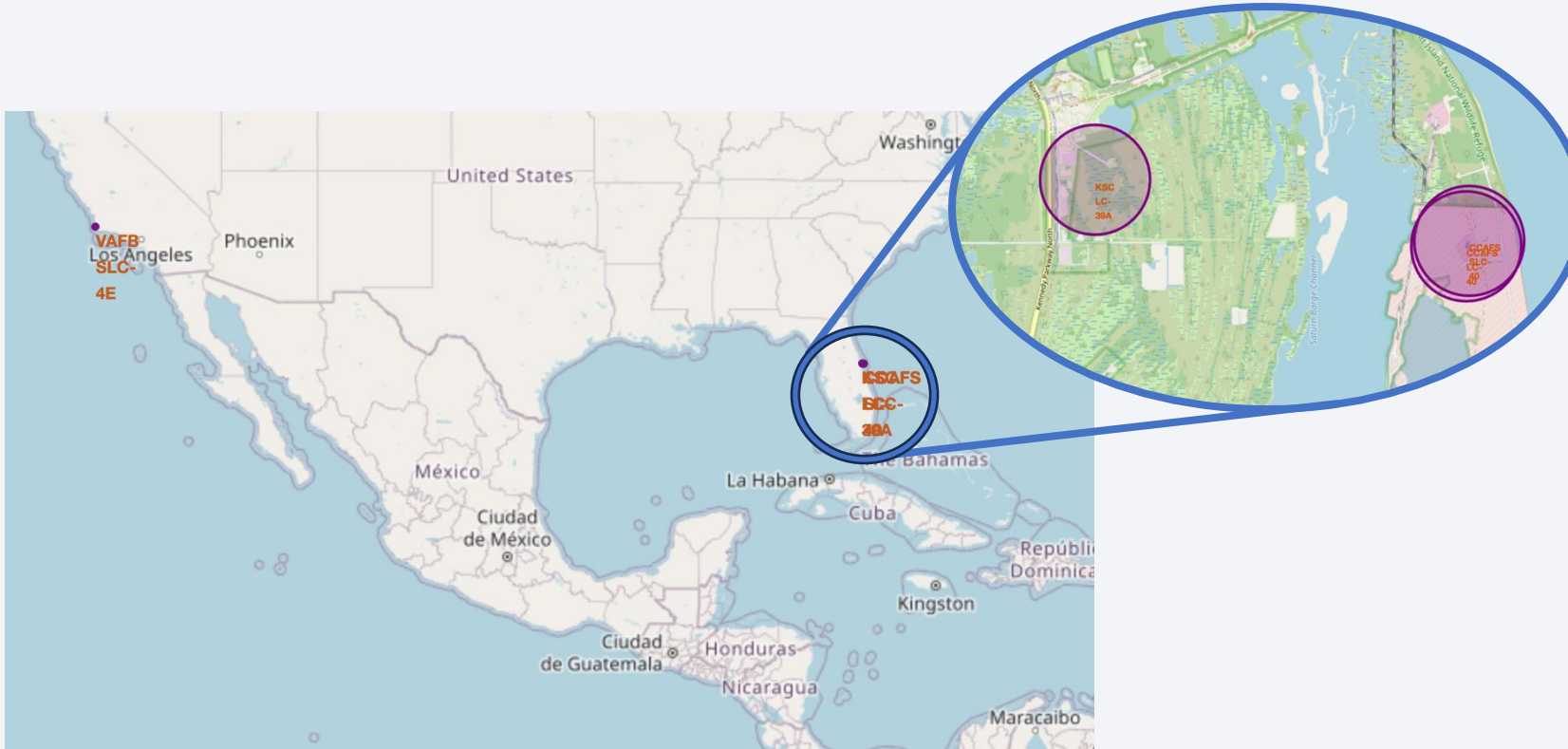
Landing outcome	Count
No attempt	9
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a curved line separating the dark surface from the deep blue of space.

Section 3

Launch Sites Proximities Analysis

Locating the launch sites on the map



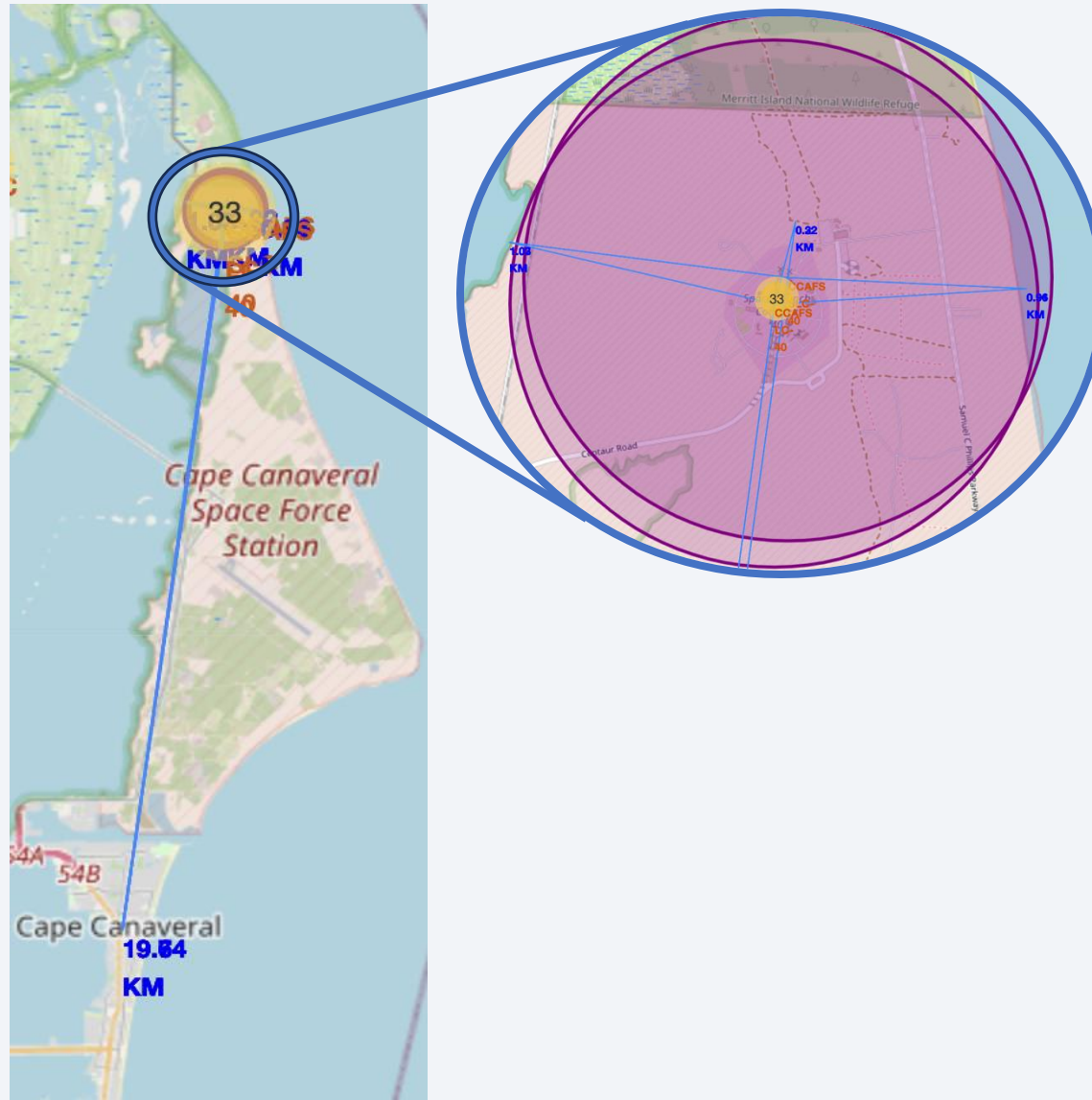
- We circled the site locations and added markers to display their name

Displaying launch outcomes

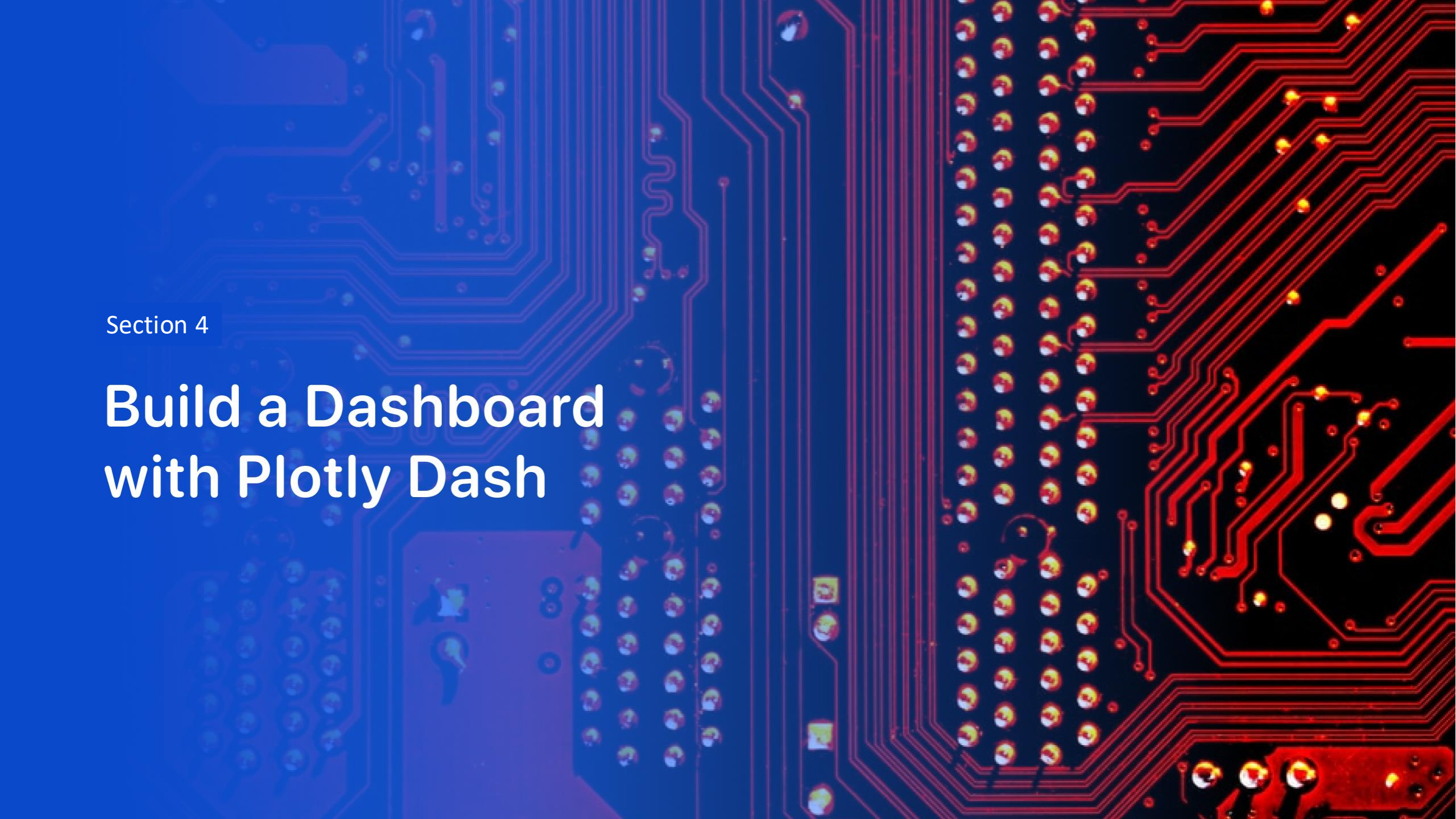


- We used a cluster of markers with their color displaying the outcome of the launch (success: green / failure: red).
- The site with the most successful launches, as seen during the EDA, is KSC-LC 39-A

Proximity analysis



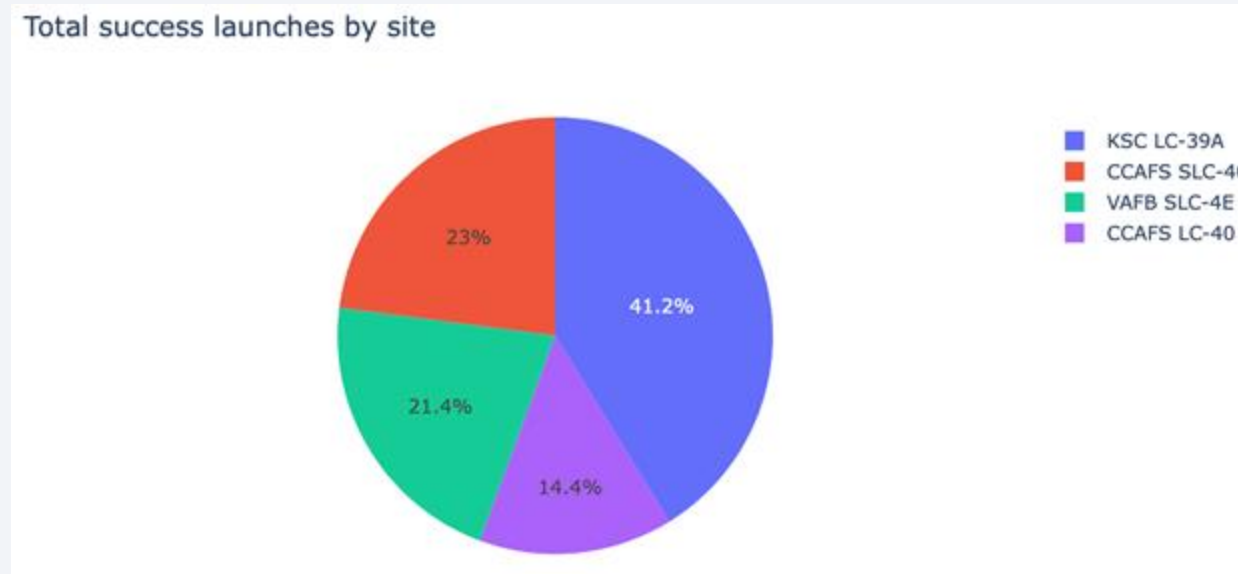
- These particular sites are situated at the same distance from the coast at both sides.
- They are also close to a rail, possibly used to transport the rockets.
- Also, they are quite far from the city of Cape Canaveral, at around 20 Km.



Section 4

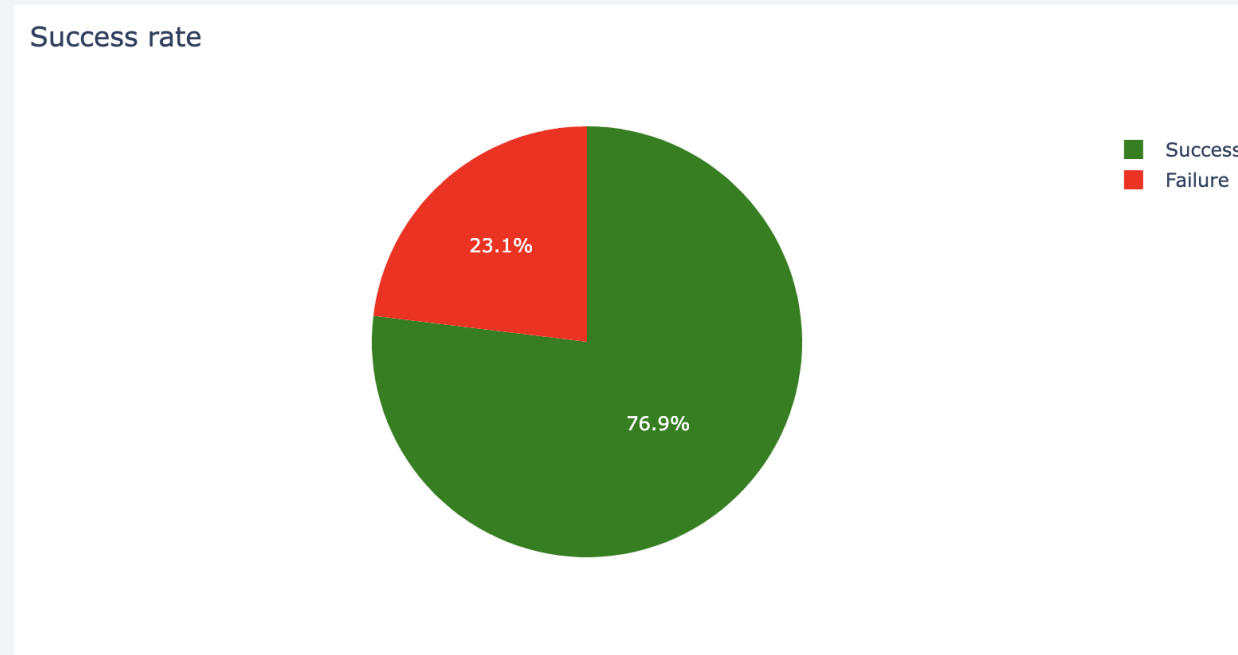
Build a Dashboard with Plotly Dash

Building a dashboard: total success rate



- Upon selecting 'All sites' in a dropdown element, in accordance with our EDA results, the most successful site is KSC LC-39A, followed by CCAFS SLC-40, VAFB SLC-4E, and CCAFS LC-40.

Building a dashboard: success rate per site



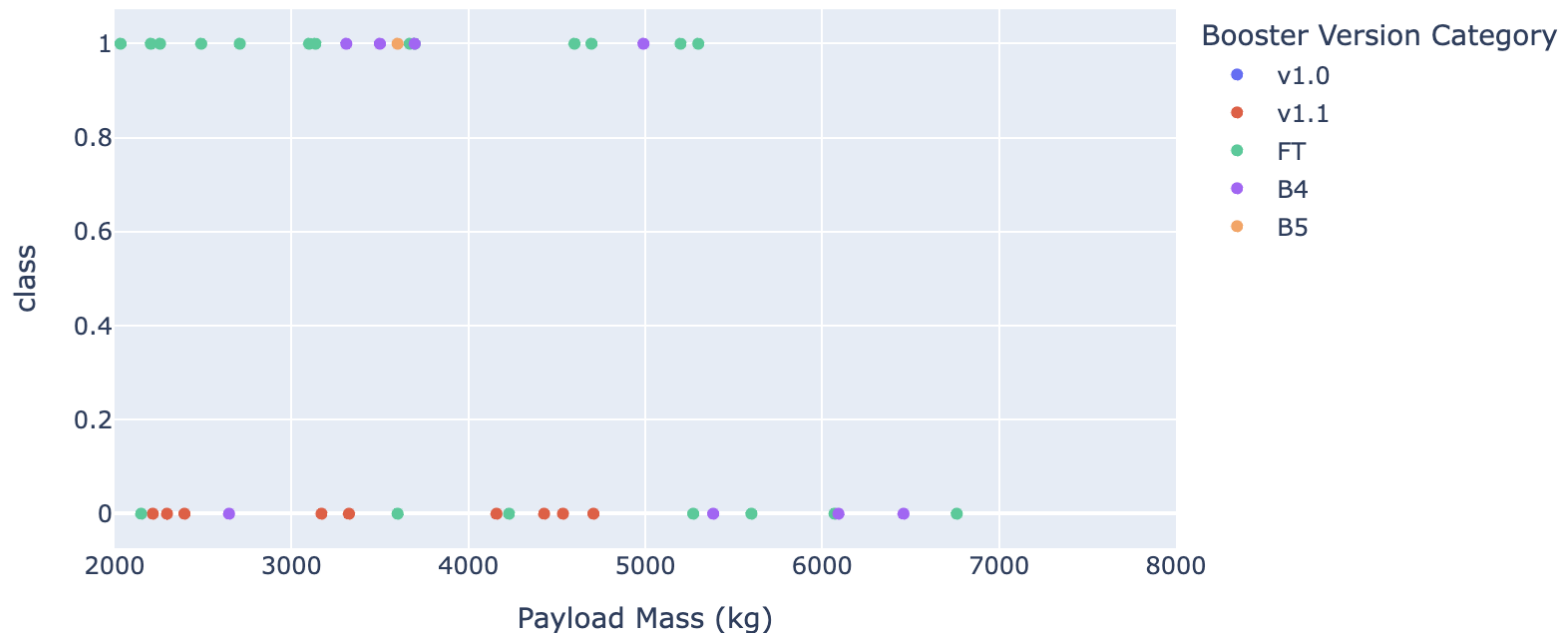
- We focus on the most successful site, KSC LC-39A. As we can see, it has a 76.9% of successful landing, and a 23.1% of failed landings.

Building a dashboard: Outcome vs payload by booster version

Payload range (Kg):



Outcome vs payload mass by booster version



- Selecting a meaningful payload range using the slider, and selecting each booster version one by one, we can see that:

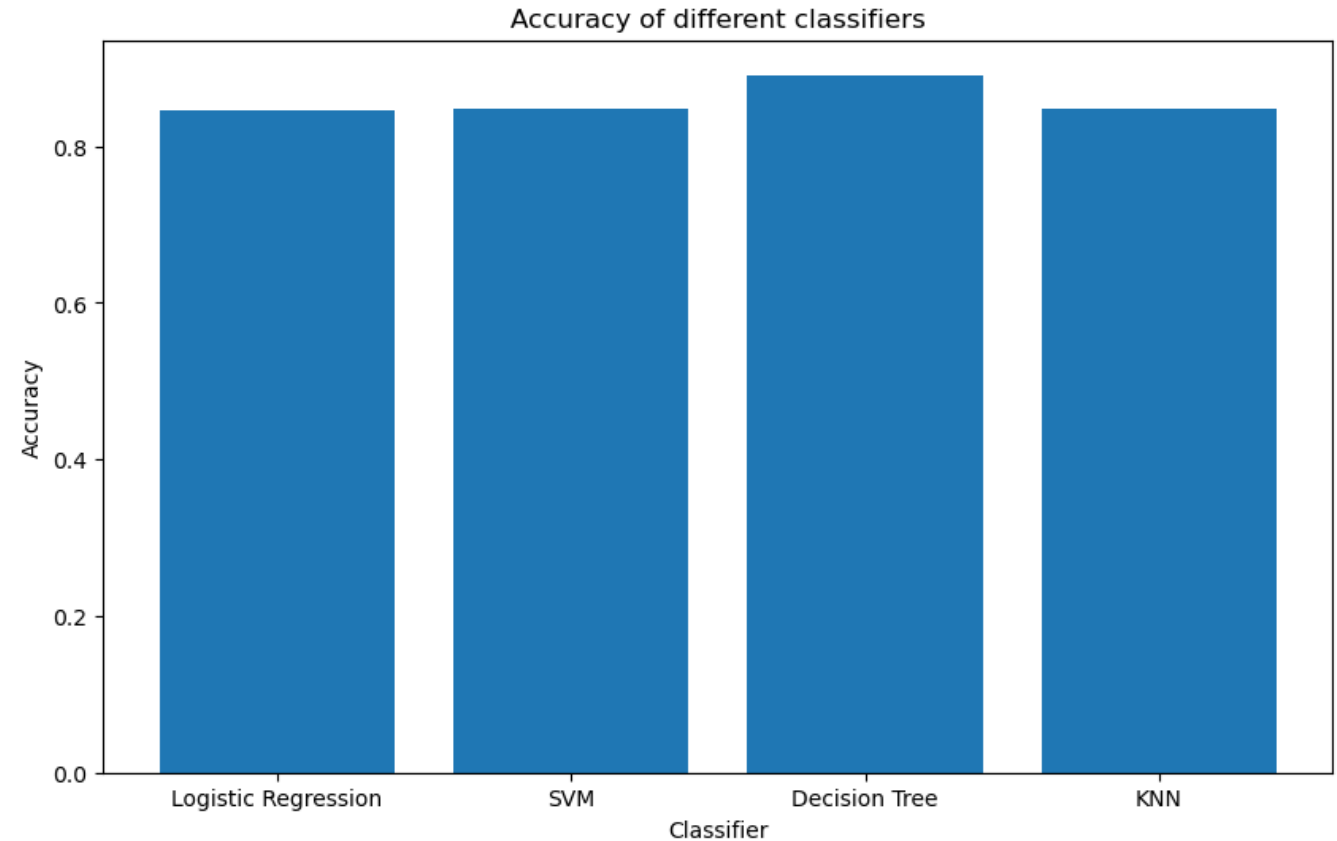
- v1.1 has only one successful outcome
- v1.0 has only failures although data terminates at 677 Kg.
- FT and B4 have slightly more successful outcomes although there does not seem to be any dependance on the payload mass.
- B5 has only one successful data point at 3600 Kg.

Section 5

Predictive Analysis (Classification)

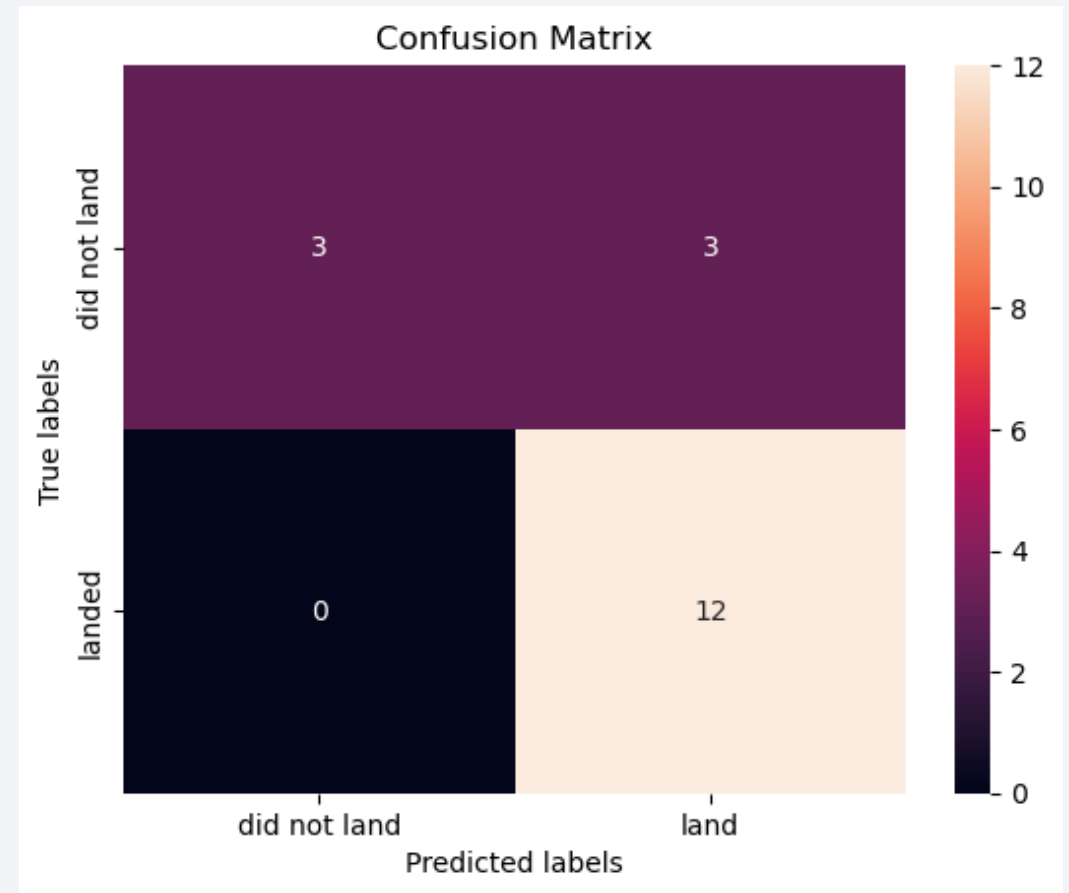
Classification Accuracy

- The Decision Tree model has a 5% higher accuracy.



Confusion Matrix

- All four models yielded the same confusion matrix, having zero false positives, and equally distributed false and true negatives.

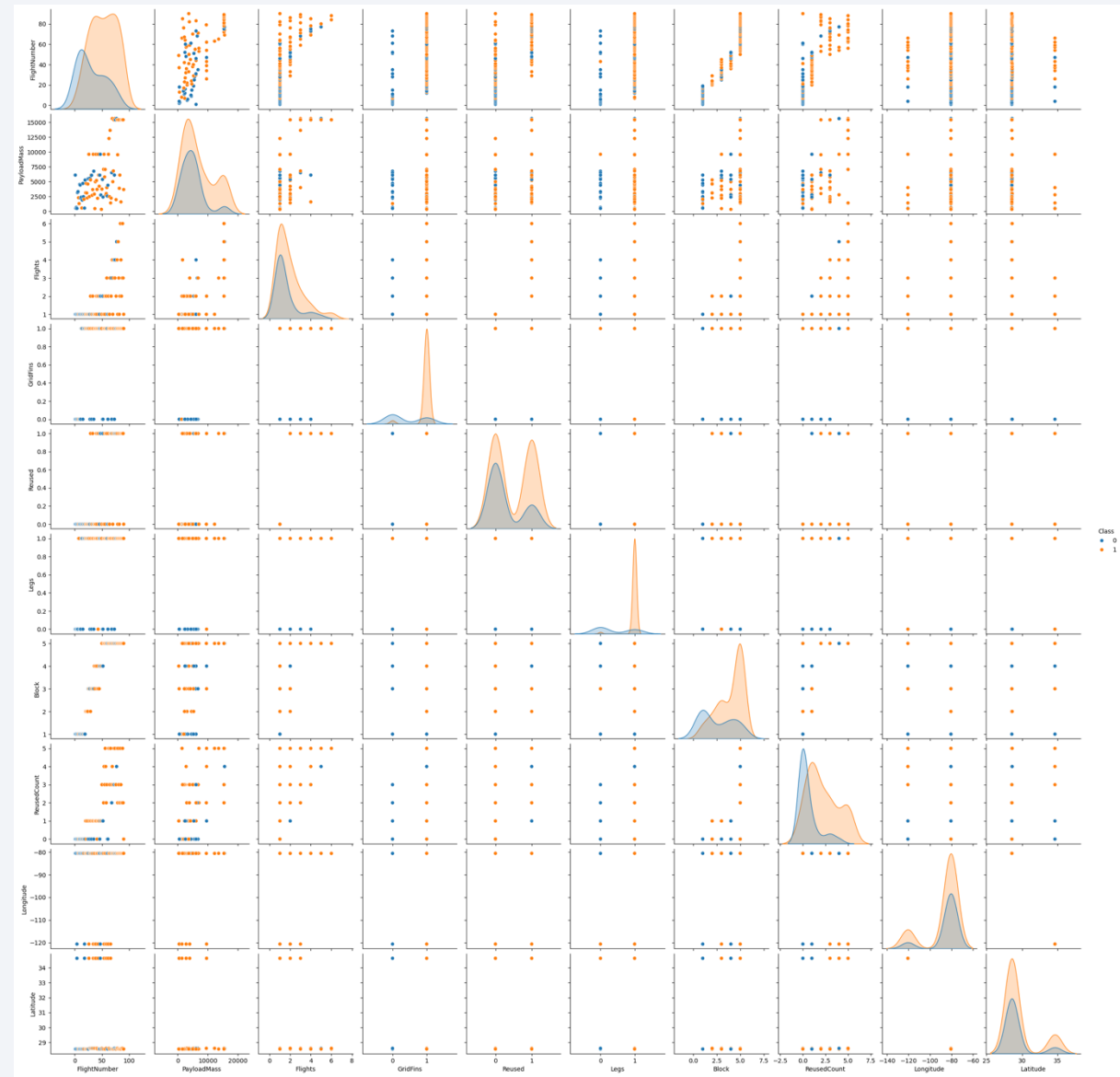


Conclusions

- We used the SpaceX API and BeautifulSoup to perform data collection.
- Performing EDA with SQL, data visualization, dash and folium revealed valuable insights about the launch locations, orbit types, payload masses, booster versions and landing pads.
- Using four different classification models, and applying grid search for hyperparameter tuning, we find the best model to be a decision tree.
- Our model can predict if a landing will be successful or not with an accuracy of 89%.

Appendix

- Crossed distributions in the dataset, classified by outcome.



Thank you!

